Privacy-Preserving and Sustainable Biomechanical Modeling: Overcoming Data Scarcity in Wearable Sensing

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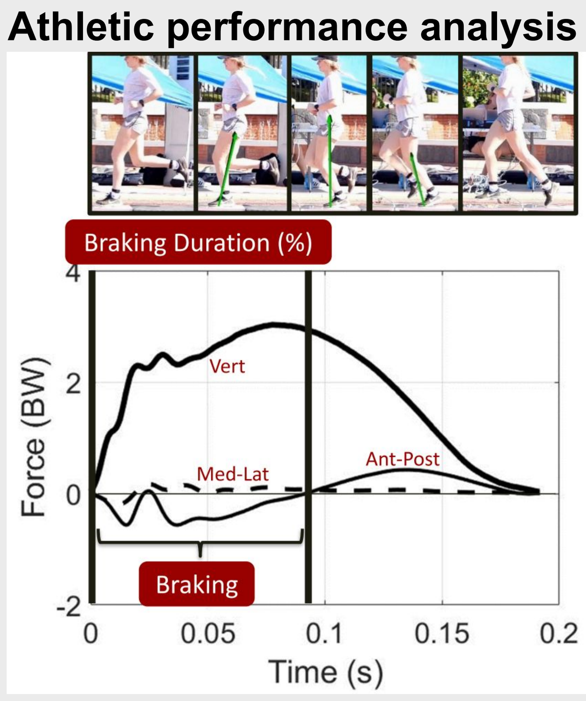
Motivation

Biomechanical measurements such as **ground reaction forces (GRFs)** and **joint moments** at the **knee** and **hip** are important because they provide insight into human movement and gait. They are used for:

Injury risk assessment

Athletic performance analysis

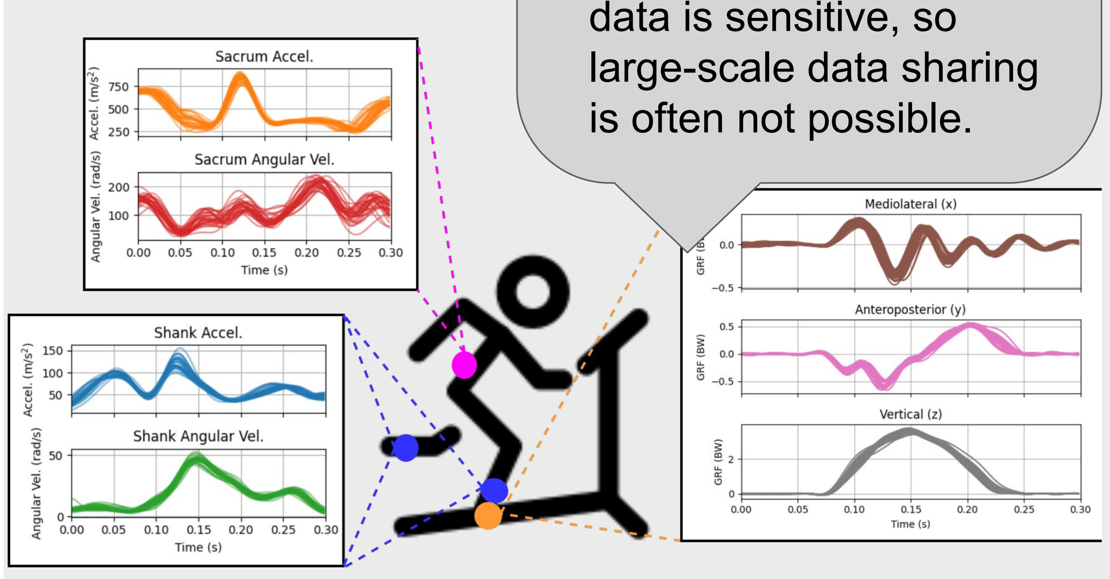


Accurate **live estimation from wearable sensors** on the **edge** could enable **applications** like:

- Real world Long term health monitoring
- Exoskeleton control for assisted movements
- Robotic systems for athletic and assistive movement

Key Problem: Lack of data

- **Data scarcity:** collecting labeled data is expensive and difficult
- **Privacy constraints:** health-related sensing data is sensitive, so large-scale data sharing is often not possible.

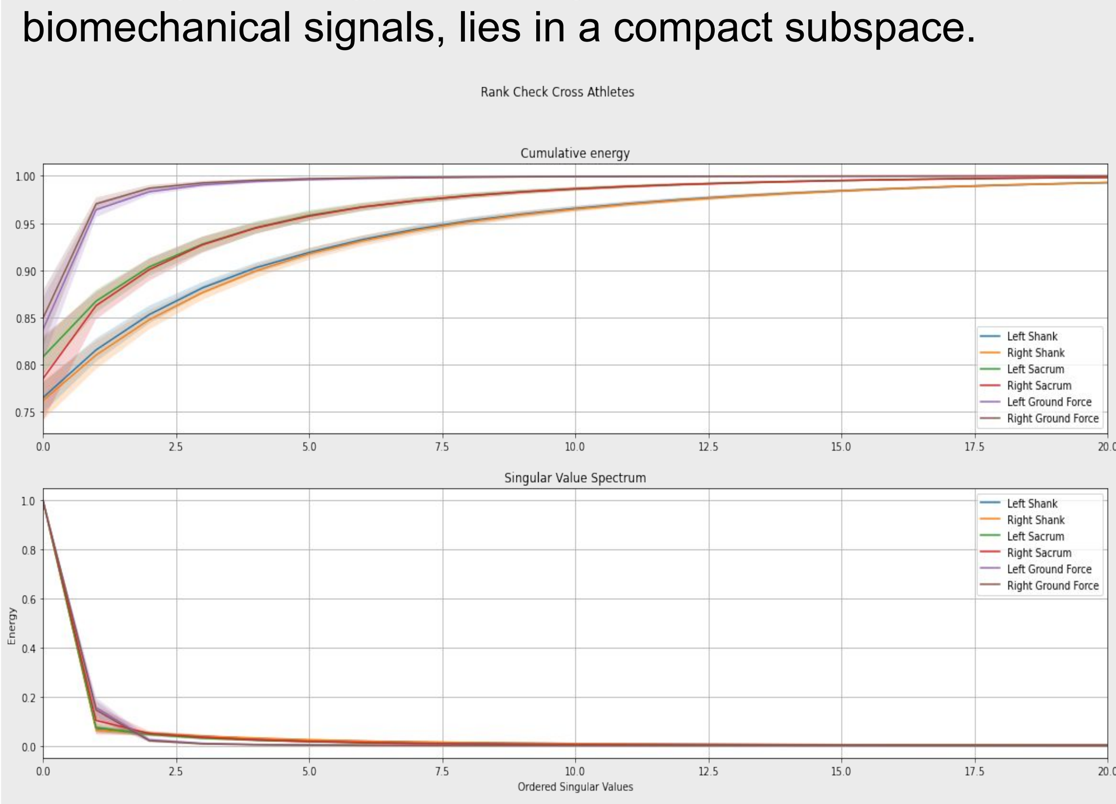


Literature trend: increasing model complexity

- Deep models dominate biomechanical estimation, including LSTMs, TCNs, and Transformers.
- Inputs often combine high-modality wearable sensing across multiple body locations, including inertial measurement units (IMUs), electromyography (EMG), and pressure insoles.
- Limited real data is often addressed through synthetic data generation, including diffusion-based augmentation.
- Some methods also generate missing modalities at inference time to handle incomplete sensor inputs.
- However, this **added complexity does not necessarily translate into better estimation accuracy, interpretability, or deployability.**

Observation: Data is Low Rank

Wearable IMU signals, including accelerometer and gyroscope measurements across body locations, exhibit strong low-rank structure. Rank **6** captures roughly **95%** of the signal energy, indicating that the dominant biomechanical signals, lies in a compact subspace.



Proposed Method

We propose SVD Embedding Regression (SER), a structured constrained low-rank regression model for biomechanical estimation.

Signal representation

For a signal matrix M, each row is one complete movement trial formed by concatenating all synchronized sensor modalities into a single vector, so each column represents the same feature across trials.

Embedding Space

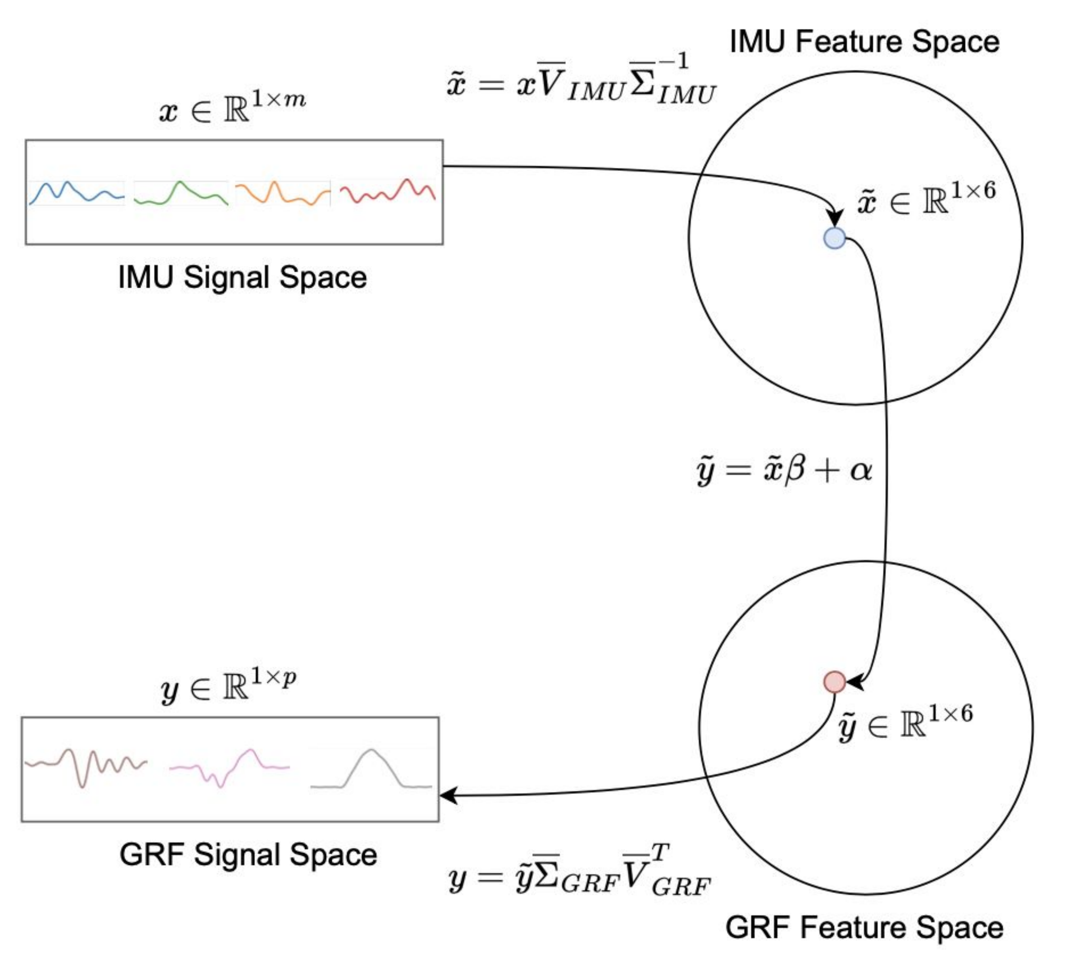
Using singular value decomposition (SVD), written as

$$M = U\Sigma V^T,$$

SER extracts a compact low-rank representation of the dominant signal structure.

Why it works

SER then learns and predicts in this constrained space, reducing noise, improving data efficiency, and helping prevent unrealistic or hallucinated biomechanical signals.



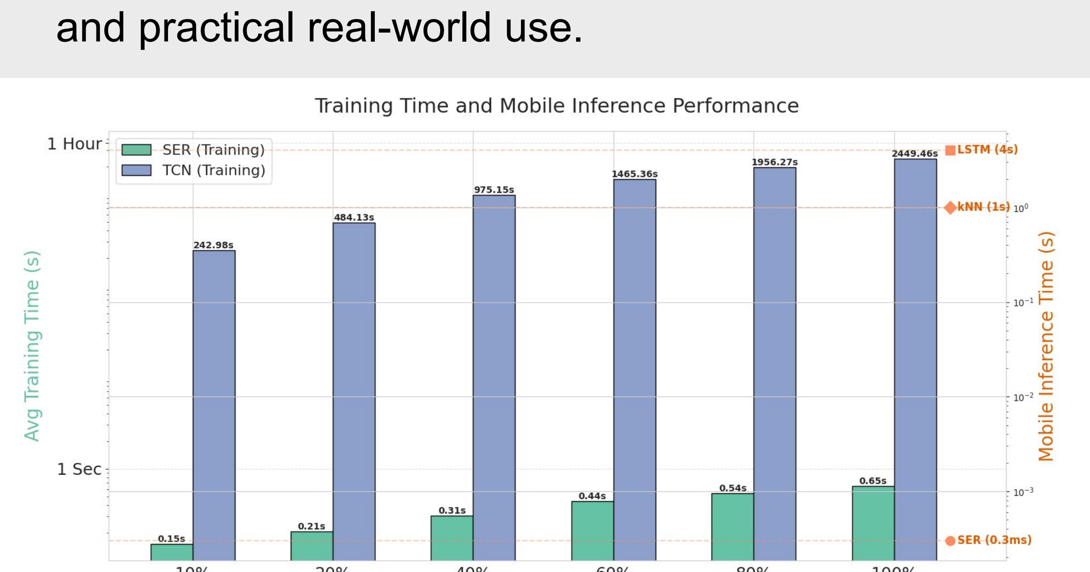
Quantitative Analysis

Model Performance and Privacy

- **Strong estimation accuracy:** Simple models like SER and KNN can outperform deep learning baselines for biomechanical estimation.
- **Robustness to data scarcity:** As training data decreases, performance degrades only slightly for simple models, whereas deep models suffer larger drops in accuracy.
- **Privacy-preserving generalization:** Personal models perform nearly as well as models trained on data from all participants, suggesting strong performance can be achieved by simple models without relying heavily on large centralized datasets.

Sustainability and Deployability

- **Lower training cost:** Simple models require substantially less training time and computation than deep learning approaches.
- **Lower deployment cost:** Simple models are more lightweight for sustainable deployment on mobile and edge devices.
- **On-device inference:** Mobile inference enables low-latency prediction while reducing cloud dependence, supporting both privacy and practical real-world use.



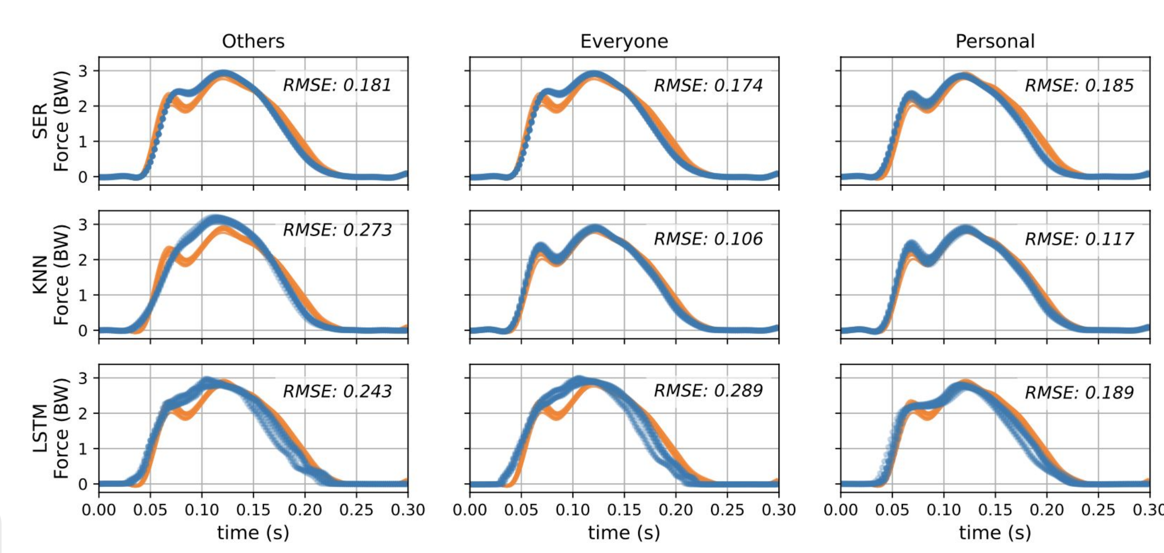
Qualitative Analysis

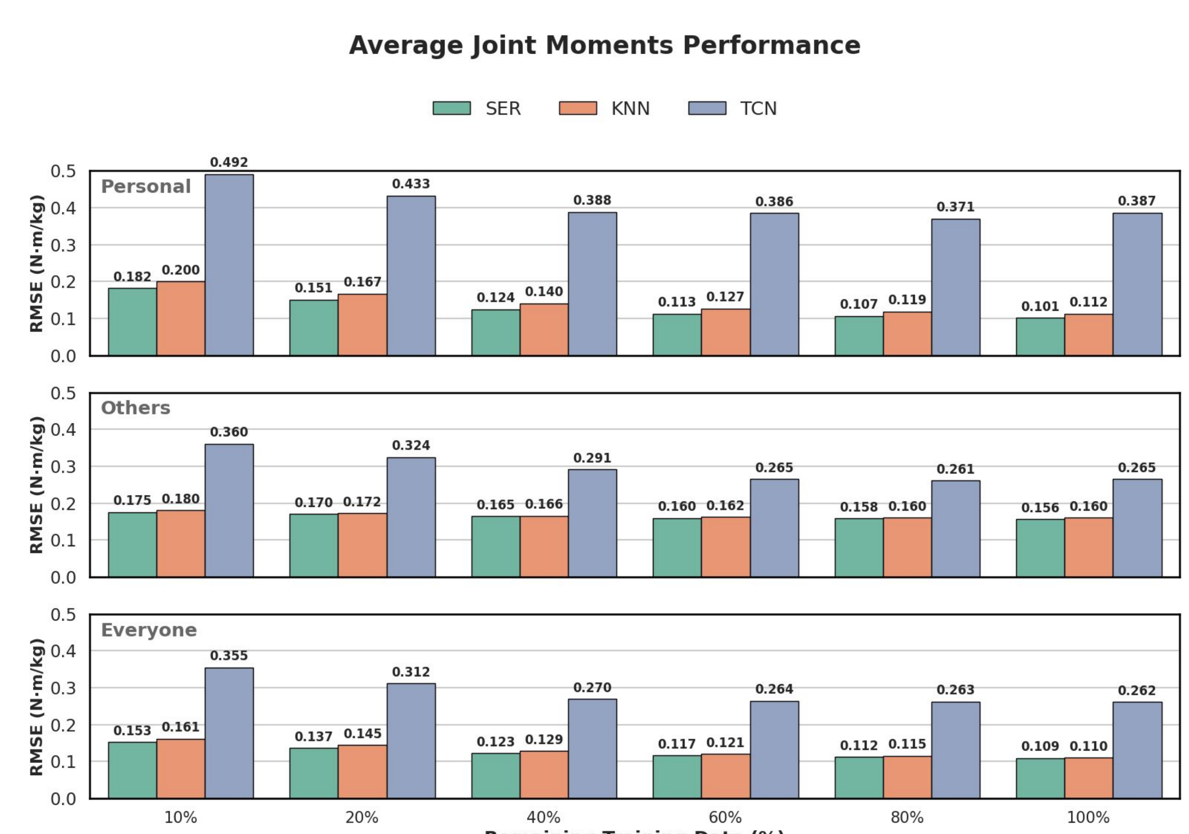
Preservation of key biomechanical structure:
Simple models better preserve biomechanical features that reflect real movement dynamics. In the example below, in most estimations, the simple model captures the vertical GRF with a clear impact peak followed by the main peak, whereas the deep model recovers only the main peak.

Better cross-subject generalization:
SER is able to recover meaningful biomechanical structure from data collected on other participants. In this example, the impact peak is preserved even without subject-specific training data.
This reduces the need for a target user to collect expensive subject-specific ground-truth data in order to obtain meaningful predictions.

Simple models predicts realistic and biomechanically plausible estimations:
Simple structured models are less likely to hallucinate unrealistic or impossible signals, producing estimates that better preserve biomechanical plausibility than deep models. For example, deep model here estimates artificial tail-end oscillation, which is inconsistent with fast running dynamics.

Personal modeling remains highly competitive:
Personal models produce estimates that are visually and qualitatively similar to models trained on data from all participants.





Conclusion

- Wearable sensing-based modeling enables scalable in-the-wild data collection while preserving privacy through accurate personal biomechanical modeling.
- Stop over-relying on complex models that require enormous unsustainable amount of resources.
- We should understand the underlying structure of the data and guide the development of interpretable models that achieve high accuracy, produce meaningful predictions, and enable impactful health and assistive applications.

Song, B., Paolieri, M., Stewart, H. E., Golubchik, L., McNitt-Gray, J. L., Misra, V., & Shah, D. (2025). Estimating ground reaction forces from inertial sensors. *IEEE Transactions on Biomedical Engineering*, *72*(2), 595-608.